Rajesh Ksheersagar

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Architect

Platform and Enterprise Architect with deep expertise in designing scalable, resilient, enterprise-grade platforms augmented with advanced AI and agentic intelligence. Over 15 years architecting mission-critical systems spanning cloud infrastructure, distributed systems, event-driven architectures, and microservices ecosystems. Specialized in integrating generative AI, LLMs, multi-agent frameworks (LangGraph), knowledge graphs, and RAG systems into enterprise platform solutions. Proven ability to bridge business strategy with technical architecture, leading cross-functional teams to deliver intelligent, autonomous platforms that solve complex enterprise challenges across healthcare, education, and SaaS.

WORK EXPERIENCE

Carelon • Hyderabad • 10/2020 - Present
Associate Architect

Skillsoft • Hyderabad • 10/2015 - 10/2020
Principal Software Engineer

Laurus Info Systems • Hyderabad • 10/2007 - 10/2015
Team Lead

PROJECTS

Claims Processing Engine

- Built an agentic AI microservice that converts healthcare Business Requirement Documents into production-ready claims test cases, using **Python, FastAPI, Pydantic v2, and Uvicorn**.
- Designed a multi-agent orchestration layer with **LangGraph and LangChain**, using LLM-based supervisor routing across Testing, Edit, Evaluation, and DevSecOps agents, with **MongoDB-backed checkpointing** for resumable workflows.
- Implemented a **RAG pipeline** for LLM grounding — document chunking, **vector embeddings** (OpenAI embeddings API), and **vector database retrieval** for context-aware test case generation, with adaptive direct-prompt fallback for small documents.
- Integrated data access through an **MCP (Model Context Protocol) tool server** over HTTP, eliminating direct database coupling; secured LLM calls with **OAuth2/JWT** and added retry/backoff via **Tenacity**.
- Delivered on **MongoDB, Redis, AWS S3/Lambda (boto3), Docker, and Jinja2**, with document parsing (**Docling, python-docx,**

SKILLS

AI & GenAI Technologies: Agentic Systems, Knowledge Graph, LangChain, LLM Integration, Ontology, Prompt Engineering

Cloud & DevOps: AWS (EMR, Lambda, Cognito, ECR), CI/CD, Docker, GKE, Kubernetes

Full Stack Development: JavaScript, Node.js, Python, React.js & Angular.js, React Native

Data Engineering & ETL: Apache Airflow, MongoDB, PostgreSQL, PySpark, Snowflake

UI/UX Development: AngularJS, Django, iOS/ Android, StoryBook, TailwindCSS, Twilio

Testing & QA: Cypress, Enzyme, Jest

Leadership: Agile Delivery, Cross-functional Team Management, Mentorship, Risk Mitigation

EDUCATION

Masters of Computer Application (MCA.) in Computer Science
Osmania University
01/2007

openpyxl), structured logging (**structlog**), and **pytest** test coverage under **SonarQube** quality gates.

- Redesign the current architecture to enable efficient and scalable parallel processing of claims
- Creating Pipeline for complete end to end processing of claims
- Integration with Splunk and Datadog for debugging
- Creating chatbot (POC) for reading BRD's and saving those into vector database(Quant)
- Development of FastAPI for exposing API for chatbot and pipeline

ICA (Intelligent Clinical Assistant)

- **Advanced OCR & Document Extraction** – Implemented dual OCR pipeline using pytesseract and AWS Tesseract for high-fidelity text extraction from medical documents, combined with PDF parsing libraries (PyPDF2, pdfplumber) to handle complex document layouts, scanned images, and structured forms with enterprise-grade accuracy.
- **Multi-Agent Orchestration with LangGraph** – Architected a sophisticated agentic framework using LangGraph to build stateful, multi-turn workflows with specialized agents for document classification, clinical data validation, and referral generation, enabling autonomous decision-making across distributed AI components.
- **Production Observability & Monitoring with Langfuse** – Integrated Langfuse for end-to-end tracing of LLM calls, agent behaviors, and pipeline metrics, enabling real-time debugging, performance analysis, cost optimization, and continuous improvement of AI-driven workflows in production healthcare environments.
- **Clinical NLP & Data Extraction** – Leveraged large language models (LLMs) with LangChain integration for intelligent entity extraction, clinical concept identification, and context-aware information synthesis from medical text, transforming unstructured documents into structured clinical datasets.
- **Scalable PDF Processing Architecture** – Designed robust PDF processing pipeline combining pytesseract, AWS services, and custom data validation layers with async processing, error handling, and quality assurance mechanisms for high-volume document ingestion while maintaining HIPAA compliance.
- **End-to-End Healthcare AI Stack** – Integrated diverse AI/ML technologies (LLMs, Computer Vision, LangGraph, Langfuse, LangChain) with healthcare workflows, demonstrating full-stack expertise in modern AI infrastructure for clinical applications.

TeleHealthOS

- Creating plans and implement feature stories.
- Taking the inputs from stakeholders and assign the tasks to team.
- Helping in solving design issues and created architecture for difficult scenarios

- Provide technical guidance and coaching to the team in software engineering best practices.
- Design and code innovative, scalable and robust features.
- Participate in all aspects of the project life cycle, from ideation to specification to release.
- Control over synchronization of the software with the system architecture

Percipio

- Architected Percipio, an enterprise e-learning mobile platform with personalized content.
- Developed microservices using Node.js and CI/CD pipelines with Jenkins.
- Adopted React Native and Redux for app performance optimization.
- Explored AI-based recommendations for learning paths.